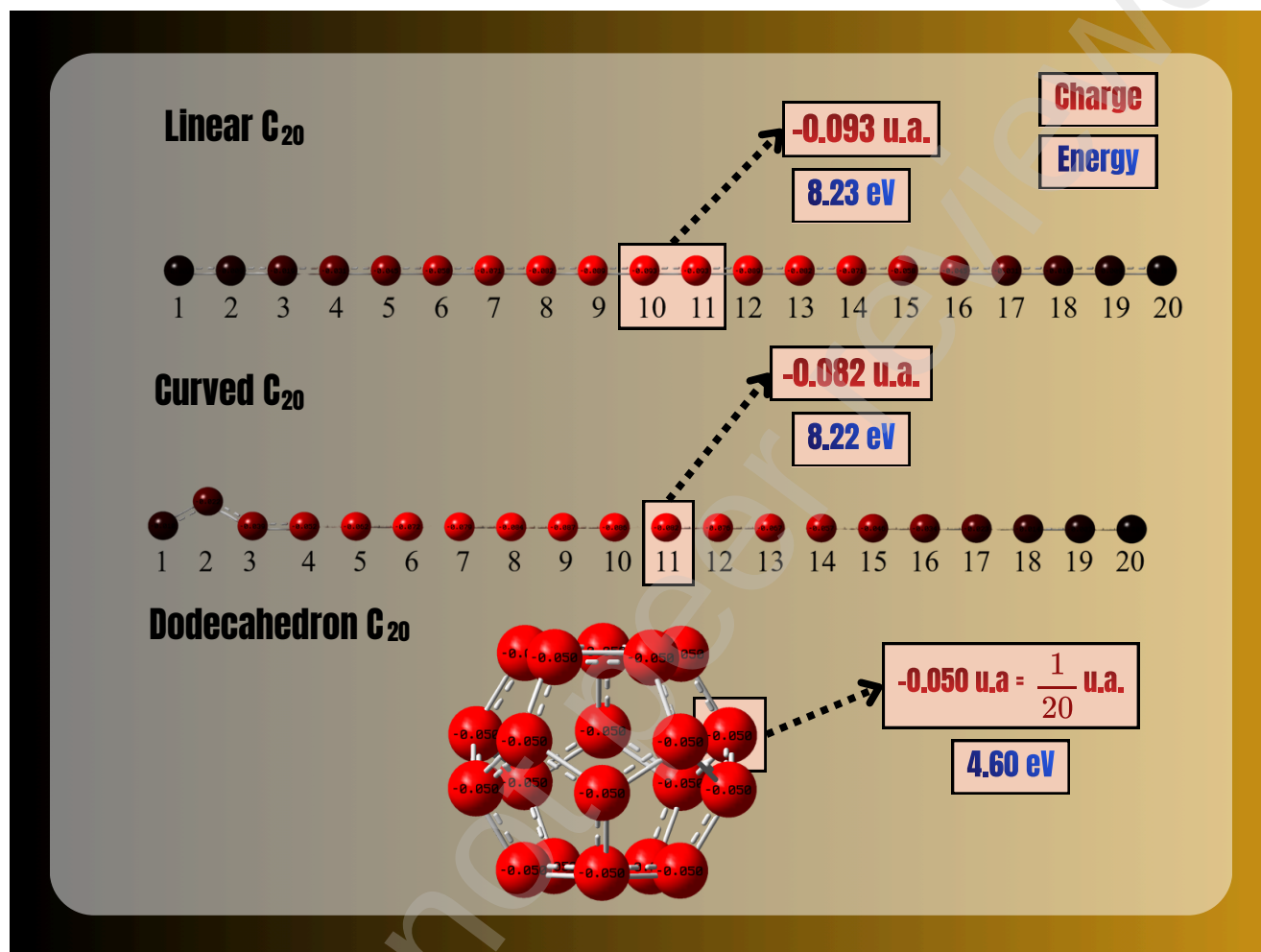


Graphical Abstract

Electron-Only Systems in Linear Chains, Curved Chains and Platonic Solids: Comparing Molecular Orbital Theory and Confinement Models

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Highlights

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- Dummy-atom molecular orbital framework accurately simulates pure electron-only confinement across 1D chains and 3D Platonic solid clusters.
- Induced curvature and polyhedral packing enhance orbital overlap, shifting electronic charge and lowering total energy through geometric potential effects.
- Single-electron energies scale inversely with electronic volume consistent with the Li-Blinder model, while multielectron systems exhibit linear increases in electron repulsion.

Electron-Only Systems in Linear Chains, Curved Chains and Platonic Solids: Comparing Molecular Orbital Theory and Confinement Models

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ARTICLE INFO

Keywords:

Platonic solids

Quantum confinement

Molecular models

Ab initio

ABSTRACT

We simulated electron-only systems using dummy atoms at the HF/LanL1MB level to evaluate molecular orbital theory against quantum confinement models. Linear systems showed expected electron localization, while introducing curvature lowered the total energy—a behavior uniquely predicted by geometric confinement theories like those of Da Costa and Mitchell. Clusters with Platonic solid and C₆₀ geometries matched Li and Blinder's model, exhibiting lower energies than linear counterparts due to enhanced orbital overlap. Furthermore, multielectron simulations revealed that total energy increases quadratically while electronic repulsion scales linearly. This approach offers a framework for future nanostructure studies.

1. Introduction

Electronic confinement can serve as a foundation for advanced investigations into transport phenomena and the electronic behavior of molecules and nanostructures. However, its theoretical description remains an active topic of research [1–8]. Different confinement models lead to distinct formulations and results for this problem [3, 4, 6, 7]. The conventional quantum confinement model, most widely used and standard in textbooks, corresponds to the direct application of the time-independent Schrödinger equation to the domain in question, where the Hamiltonian is reduced in practice to the Laplacian operator [1, 4].

Alternative confinement models have also been proposed and utilized; in these frameworks, in addition to the Laplacian operator, the Schrödinger equation incorporates geometric operators that depend on the curvature of the space [2, 3, 5–9]. There is still no consensus in the literature regarding which formulation is the most appropriate for confinement. A widely adopted approach for confinement to curves and surfaces is Da Costa's model [2], which introduces an exclusively attractive geometric potential dependent on principal curvatures. Mitchell later established a new formulation [9] that generalized Da Costa's results to arbitrary manifolds, incorporating additional operators into the Hamiltonian, such as geodesic curvature, and introducing the possibility of a repulsive geometric potential. On a sphere, for instance, Da Costa's geometric potential vanishes, whereas Mitchell's depends on geodesic curvatures. These differing formulations provide both the foundation and the motivation for new theoretical and computational studies.

Platonic solids have long captivated scientific curiosity. Originating from mathematical philosophy, these regular

polyhedra are also ubiquitous in nature [10]. Diverse systems—including molecules, crystals, and lanthanide complexes—rely on these geometric arrangements for their structural stability [11, 12]. Li and Blinder investigated the quantum behavior of a particle confined within the five Platonic solids (tetrahedron, octahedron, hexahedron, icosahedron, and dodecahedron) as well as within a sphere, deriving an approximate analytical expression for their ground-state energies [13].

In this work, we investigate purely electronic systems using Molecular Orbital Theory and compare the results with both conventional confinement models and those involving spatial curvature. We leverage the capability in Computational Quantum Chemistry of employing dummy atoms, centers hosting only wave functions [18], to model systems composed solely of electrons and evaluate them against theoretical confinement predictions. Specifically, we study linear chains of dummy atoms, chains with curved sections, and clusters arranged in Platonic solid geometries.

2. Computational Details

We constructed molecular models using dummy atoms, i.e., centers consisting solely of wave functions. We built linear chains as well as chains with a curved section, with the number of centers corresponding to the number of vertices of the five Platonic solids: tetrahedron (T), octahedron (O), hexahedron (H), icosahedron (I), and dodecahedron (D). We also prepared sixty-center linear systems to compare with the C₆₀ fullerene sphere. The specific angles of each geometry were preserved, and we adopted a typical carbon–carbon bond distance of 1.42 Å. The curved sections were introduced at the second dummy atom of the chain, tilted at an angle of 150° relative to the rest of the chain (Fig. 2).

Computational calculations were performed at the Hartree–Fock (HF) level [14] (Eq. (1)) using the LanL1MB basis set [15–17] (Eq. (2)), as implemented in the Gaussian software package [18].

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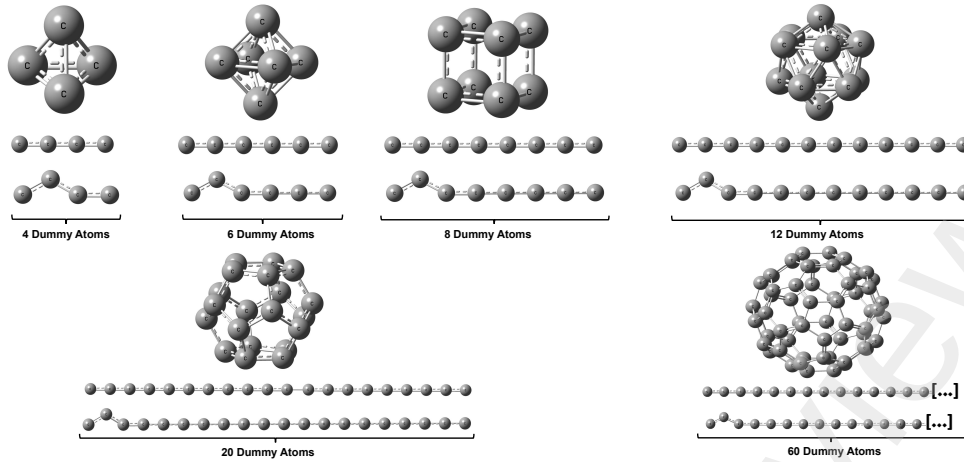


Figure 1: Molecular models for the Platonic solids and their linear counterparts.

$$E = 2 \sum_i H_i + \sum_{ij} (2J_{ij} - K_{ij}). \quad (1)$$

$$\Phi_l = \sum_i C_i r^l e^{-\alpha_i r^2} \quad (2)$$

where J_{ij} and K_{ij} are the Coulomb and exchange integrals, respectively, H_i represents the one-electron core Hamiltonian energy, C_i is the contraction coefficient for the i -th Gaussian, r^l is the radius associated with angular momentum, α_i is the exponent (orbital exponent parameter), and r is the radial distance.

Computational calculations were carried out considering one or more electrons. In this manner, for systems with two or more interacting electrons, we can potentially achieve a more realistic description of multiple free electrons confined within a region by directly observing electronic repulsion effects—a scenario that presents greater analytical complexity for traditional confinement models.

For single-electron solids, we compared our results with the approximate analytical model proposed by Li and Blinder (Eq. (3)):

$$E_0(n_v) \approx \left(12.82 + \frac{38.62}{n_v^{1.429}} \right) \frac{\hbar^2}{mV^{2/3}} \quad (3)$$

where n_v is the number of vertices, $\hbar$ is the reduced Planck constant, V is the volume of the solid, and m represents the mass of the particle.

3. Results and Discussion

All calculations, involving one or more electrons, successfully converged. The single-electron linear chains of dummy atoms with length L exhibited the highest charge concentration around the center of the chain, $L/2$, and the total energy decreased as the linear chain length increased.

This behavior aligns with predictions from both conventional [1, 4] and geometric [2, 3, 7] confinement models. In all purely linear systems, the electron exhibited higher kinetic energy compared to the corresponding systems featuring curved sections. In contrast, the electronic charge distribution is attracted toward the curved regions. This finding finds an analog exclusively in geometric confinement models, such as those of Da Costa [2] and Mitchell [9]. The conventional model fails in this prediction, as it still indicates charge localization around $L/2$.

In the case of a sphere (approximated here by C_{60}) compared with the line, the energy decreased from 8.19 eV to 4.83 eV, roughly by half. In our interpretation, a higher packing of orbitals occurs due to the spatial proximity of the atoms imposed by the geometry. For a sphere, Da Costa's geometric potential vanishes; hence, this model provides no insight into the energy shift in this case. On the other hand, in Mitchell's formulation, the sphere possesses a geometry-induced attractive potential, which implies that the electron in this description should indeed have lower energy than in a straight line.

The energies obtained for the single-electron solids followed the predictions of Li and Blinder. We also observed an analogous behavior between the volume of the solid and the electronic volume (V_{elec}) with respect to energy. The electron energies in the solids were lower than those in the corresponding linear systems with the same number of dummy atoms. From the perspective of Molecular Orbital Theory, this can be interpreted as greater orbital overlap arising from the close packing of dummy atoms in the polyhedra, rendering the systems more stable and lower in energy relative to their linear counterparts.

Table 1 presents the single-electron energies in the Platonic solids obtained from Li and Blinder's model (E_{LB}) alongside our HF/LanL1MB calculations for systems with one and multiple electrons.

For single-electron systems, the calculated energies were very close to those predicted by the referenced analytical model. The electronic volume obtained from the calculations

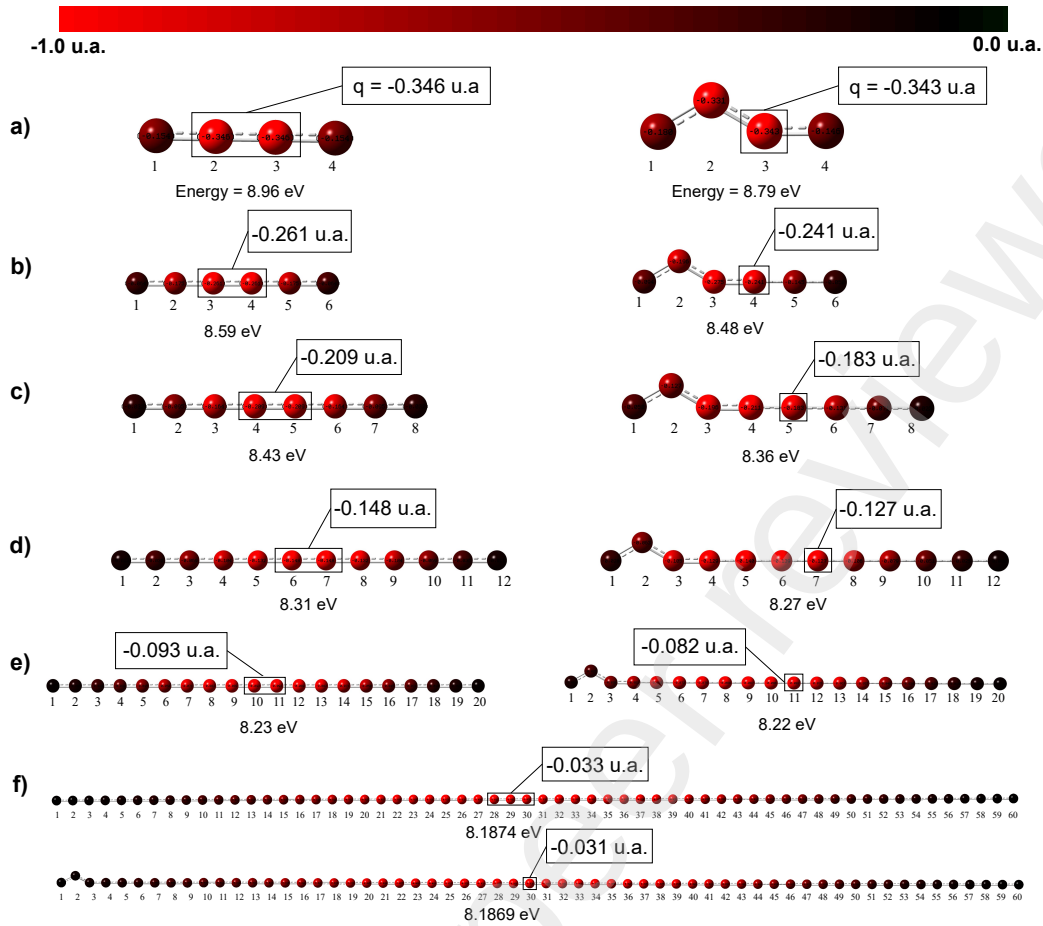


Figure 2: Charge distributions and values (q), in a.u., and energies, in electron-volts, for the linear and curved-section system models from (a) to (f) with 4, 6, 8, 12, 20, and 60 centers, respectively. The number of centers corresponds to the number of vertices of the Platonic solids and the sphere approximated by C_{60} .

Table 1

Energies for a single electron confined in the Li–Blinder (LB) model and for one, two, and three electrons obtained from HF/LanL1MB calculations.

Solid	$V_{\text{elct}} (\text{\AA}^3)$	n_v	E_{LB} (eV)	$E_{HF, 1e}$ (eV)	$E_{HF, 2e}$ (eV)	$E_{HF, 3e}$ (eV)	$E_{HF, 4e}$ (eV)	$E_{HF, 5e}$ (eV)
T	55.345	4	9.523	7.765	27.453	63.377	110.038	163.941
O	76.880	6	6.659	6.507	23.904	55.612	98.056	147.208
H	104.288	8	5.092	5.688	21.127	48.831	86.592	126.903
I	120.922	12	4.338	4.829	18.623	43.340	76.961	115.880
D	186.672	20	3.115	4.603	15.835	34.313	59.906	89.452

is consistent with the predictions of Li and Blinder, revealing an inversely proportional relationship to the energies scaling approximately as V^{-1} . A major advantage of applying Molecular Orbital Theory (MOT) is the capability to perform multi-electron calculations. We observed that the total energies increase almost quadratically with the number of electrons. Power-law fits of the form $y = ax^b$ yielded coefficients of determination (R^2) close to 1, with the exponent b ranging between 1.89 and 1.93. Linear fits of the electronic repulsion energy as a function of the number of electrons also resulted in an R^2 very close to 1.

4. Conclusion

By calculating dummy-atom molecular models via Molecular Orbital Theory (MOT) at the Hartree–Fock/LanL1MB level, we were able to exclusively simulate one or more electrons in linear chains, linear chains with curved sections, and clusters arranged in Platonic solid geometries, comparing the results with quantum confinement models. For a single electron in linear chains of length L , the results reproduce the predictions of confinement models: the energy decreases with increasing chain length, and the charge is concentrated around $L/2$. When a curved section is introduced into the

linear system, it attracts the electron and lowers the total energy compared to the purely linear chain. This result finds an analog exclusively in geometric confinement models, such as those of Da Costa and Mitchell. From the perspective of MOT, we propose that the curved sections induce greater orbital overlap, thereby stabilizing the system.

The results obtained for a single electron in the Platonic solid clusters showed good agreement with the predictions of the Li and Blinder model, including the calculated energies and the electronic volume, which exhibits an inversely proportional relationship to energy, mirroring the behavior of the geometric volume. Single-electron energies in the polyhedra were lower than those obtained for the corresponding linear chains with the same number of dummy atoms. According to our interpretation, greater orbital interaction occurs in the solids due to the higher spatial packing of the atoms, leading to greater system stability. The energy obtained for a single electron in the dummy C_{60} cluster was lower than in the sixty-center linear chain. In Da Costa's theoretical model, the geometric potential of a sphere is zero, which offers no explanation for this observation. In contrast, Mitchell's model attributes a geometric potential to the sphere due to geodesic curvatures, providing a consistent analog for the lower energy obtained.

In the Platonic solid clusters containing multiple electrons, the electronic repulsion energy was found to increase linearly with the number of electrons. This work paves the way for future investigations involving single or multiple electrons confined in systems of diverse geometries, such as nanostructures and molecular electronic devices.

Acknowledgements

We acknowledge the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq).

CRedit authorship contribution statement

Bruna M. G. Oliveira: Computational calculations, Data curation, Writing - Original draft preparation. **Eduardo V. S. Anjos:** Computational calculations, Data curation, Writing - Original draft preparation. **Cristiano C. Bastos:** Data curation, Data analysis, Writing - Original draft preparation, Writing - final version, Funding acquisition.

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